Ideation Phase

Brainstorm & Idea Prioritization Template

Date	10 July 2026
Team ID	
Project Name	Smart Lender – AI-Powered Loan Approval Prediction System
Maximum Marks	4 Marks

Purpose

Brainstorming provides a free and open environment that encourages creative thinking to arrive at the ideal solution. This session was used to explore approaches, group related ideas, and prioritize the ones that best solve the loan approval problem.

Problem Statement

Predict whether a bank loan application should be approved or rejected based on applicant details, so that the process is faster, more consistent, and less dependent on manual judgement.

Idea Generation

- Rule-based eligibility checklist (rejected – too rigid, cannot capture non-linear patterns in applicant data).
- Classical ML classification models: Decision Tree, Random Forest, K-Nearest Neighbors (KNN), XGBoost.
- Flask-based web application for data entry and instant results.
- Business-rule layer on top of the model to translate a raw prediction into eligibility percentage, confidence score, risk level, reasons and recommendations.
- Applicant profile and history pages so users can review past predictions.

Grouping of Ideas

Category	Ideas Grouped Under It
Data Handling	Missing-value imputation, label encoding, feature scaling
Modeling	Decision Tree, Random Forest, KNN, XGBoost – compare and select the best
Deployment	Flask web app, saved model/encoder/scaler (.pkl) artifacts
User Experience	Predict form, Result page with reasons & recommendations, History page

Prioritization

- Priority 1: Clean and preprocess the loan_data.csv dataset (614 records).
- Priority 2: Train and compare Decision Tree, Random Forest, KNN and XGBoost.
- Priority 3: Select the best-performing model (KNN, 84.55% accuracy) and save model/encoder/scaler.
- Priority 4: Build the Flask application (Home, Predict, Result, About, Applicant Profile, History pages).

- Priority 5: Add eligibility percentage, confidence score, risk level, reasons and recommendations via business logic on top of the prediction probability.